Modul 3
Matematika Diskrit Graf
Konektivitas Graf





105 – Ivanka Bunga Aulia– 26/10/2026

123 – Syadida Tsaqifa Nada – 26/10/2026

124 – Zahra Azkiya Rahmani – 26/10/2026

Contents

Tujuan Pembelajaran 3

Materi Esensial 3

Materi untuk Praktikum 3

Definisi 3

Koneksi Undirected Graf 6

Koneksi Directed Graf 10

Tutorial Praktikum 11

Tutorial01.py 11

Task Praktikum 12

Task Individu 12

Task01_Individu.py 12

Task02_Individu.py ..... **Error! Bookmark not defined.**

Task Kelompok (Soal Kolaboratif & Sharing) 14

Task03_Timproyek.py 14

Task04_FullMesh.py 17

Task05_Datasister.py 18

Tujuan Pembelajaran

- Mahasiswa mampu mengidentifikasi dan memvisualisasikan path dan circuit pada sebuah graf.
- Mahasiswa mampu menentukan connectivity sebuah graf (terhubung, tidak terhubung, terhubung kuat, atau terhubung lemah).
- Mahasiswa mampu mengidentifikasi cut vertex dan cut edge sebagai titik kritis dalam sebuah jaringan.
- Mahasiswa mampu menggunakan NetworkX untuk menganalisis keterhubungan graf dan menyelesaikan studi kasus.

Materi Esensial

Konektifitas Graf:

- **Path (Lintasan):** Urutan simpul dan sisi yang menghubungkan dua simpul.
- **Circuit (Sirkuit):** Path yang berawal dan berakhir di simpul yang sama.
- **Connected Graph (Graf Terhubung):** Graf tak berarah di mana terdapat path antara setiap pasangan simpul.
- **Connected Components (Komponen Terhubung):** Subgraf terhubung maksimal dari suatu graf.
- Cut Vertex (Titik Potong) / Articulation Point: Simpul yang, jika dihapus, akan membuat graf menjadi tidak terhubung atau meningkatkan jumlah komponen terhubung.
- Cut Edge (Sisi Potong) / Bridge: Sisi yang, jika dihapus, akan membuat graf menjadi tidak terhubung atau meningkatkan jumlah komponen terhubung.

Properti Dasar:

- Strongly Connected (Terhubung Kuat): Setiap simpul dapat dicapai dari simpul lain melalui directed path.
- Weakly Connected (Terhubung Lemah): Underlying undirected graph (graf dengan arah sisi diabaikan) adalah graf terhubung.
- Strongly Connected Components: Subgraf terhubung kuat maksimal dalam sebuah graf berarah.

Materi untuk Praktikum

Definisi

10.4.1 Introduction

Many problems can be modeled with paths formed by traveling along the edges of graphs. For instance, the problem of determining whether a message can be sent between two computers using intermediate links can be studied with a graph model. Problems of efficiently planning routes for mail delivery, garbage pickup, diagnostics in computer networks, and so on can be solved using models that involve paths in graphs.

10.4.2 Paths

Informally, a **path** is a sequence of edges that begins at a vertex of a graph and travels from vertex to vertex along edges of the graph. As the path travels along its edges, it visits the vertices along this path, that is, the endpoints of these edges.

A formal definition of paths and related terminology is given in Definition 1.

Definition 1

Let n be a nonnegative integer and G an undirected graph. A *path* of *length* n from u to v in G is a sequence of n edges $e_1, \ldots, e_n$ of G for which there exists a sequence $x_0 = u, x_1, \ldots, x_{n-1}, x_n = v$ of vertices such that e_i has, for $i = 1, \ldots, n$, the endpoints x_{i-1} and x_i. When the graph is simple, we denote this path by its vertex sequence $x_0, x_1, \ldots, x_n$ (because listing these vertices uniquely determines the path). The path is a *circuit* if it begins and ends at the same vertex, that is, if $u = v$, and has length greater than zero. The path or circuit is said to *pass through* the vertices $x_1, x_2, \ldots, x_{n-1}$ or *traverse* the edges $e_1, e_2, \ldots, e_n$. A path or circuit is *simple* if it does not contain the same edge more than once.

When it is not necessary to distinguish between multiple edges, we will denote a path $e_1, e_2, \ldots, e_n$, where e_i is associated with $\{x_{i-1}, x_i\}$ for $i = 1, 2, \ldots, n$ by its vertex sequence $x_0, x_1, \ldots, x_n$. This notation identifies a path only as far as which vertices it passes through. Consequently, it does not specify a unique path when there is more than one path that passes through this sequence of vertices, which will happen if and only if there are multiple edges between some successive vertices in the list. Note that a path of length zero consists of a single vertex.

Remark: There is considerable variation of terminology concerning the concepts defined in Definition 1. For instance, in some books, the term **walk** is used instead of *path*, where a walk is defined to be an alternating sequence of vertices and edges of a graph, $v_0, e_1, v_1, e_2, \ldots, v_{n-1}, e_n, v_n$, where v_{i-1} and v_i are the endpoints of e_i for $i = 1, 2, \ldots, n$. When this terminology is used, **closed walk** is used instead of *circuit* to indicate a walk that begins and ends at the same vertex, and **trail** is used to denote a walk that has no repeated edge (replacing the term *simple path*). When this terminology is used, the terminology **path** is often used for a trail with no repeated vertices, conflicting with the terminology in Definition 1. Because of this variation in terminology, you will need to make sure which set of definitions are used in a

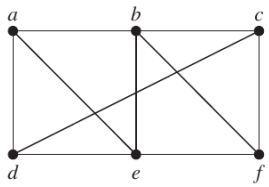


FIGURE 1 **A simple graph.**

EXAMPLE 1 In the simple graph shown in Figure 1, a, d, c, f, e is a simple path of length 4, because $\{a, d\}$, $\{d, c\}$, $\{c, f\}$, and $\{f, e\}$ are all edges. However, d, e, c, a is not a path, because $\{e, c\}$ is not an edge. Note that b, c, f, e, b is a circuit of length 4 because $\{b, c\}$, $\{c, f\}$, $\{f, e\}$, and $\{e, b\}$ are edges, and this path begins and ends at b. The path a, b, e, d, a, b, which is of length 5, is not simple because it contains the edge $\{a, b\}$ twice. ◀

Definition 2

Let n be a nonnegative integer and G a directed graph. A *path* of length n from u to v in G is a sequence of edges $e_1, e_2, \ldots, e_n$ of G such that e_1 is associated with (x_0, x_1), e_2 is associated with (x_1, x_2), and so on, with e_n associated with (x_{n-1}, x_n), where $x_0 = u$ and $x_n = v$. When there are no multiple edges in the directed graph, this path is denoted by its vertex sequence $x_0, x_1, x_2, \ldots, x_n$. A path of length greater than zero that begins and ends at the same vertex is called a *circuit* or *cycle*. A path or circuit is called *simple* if it does not contain the same edge more than once.

Remark: Terminology other than that given in Definition 2 is often used for the concepts defined there. In particular, the alternative terminology that uses *walk, closed walk, trail*, and *path* (described in the remarks following Definition 1) may be used for directed graphs. See [GrYe05] for details.

Note that the terminal vertex of an edge in a path is the initial vertex of the next edge in the path. When it is not necessary to distinguish between multiple edges, we will denote a path $e_1, e_2, \ldots, e_n$, where e_i is associated with (x_{i-1}, x_i) for $i = 1, 2, \ldots, n$, by its vertex sequence $x_0, x_1, \ldots, x_n$. The notation identifies a path only as far as which the vertices it passes through. There may be more than one path that passes through this sequence of vertices, which will happen if and only if there are multiple edges between two successive vertices in the list.

Paths represent useful information in many graph models, as Examples 2–4 demonstrate.

EXAMPLE 2

Links

Paths in Acquaintanceship Graphs In an acquaintanceship graph there is a path between two people if there is a chain of people linking these people, where two people adjacent in the chain know one another. For example, in Figure 6 in Section 10.1, there is a chain of six people linking Kamini and Ching. Many social scientists have conjectured that almost every pair of people in the world are linked by a small chain of people, perhaps containing just five or fewer people. This would mean that almost every pair of vertices in the acquaintanceship graph containing all people in the world is linked by a path of length not exceeding four. The play *Six Degrees of Separation* by John Guare is based on this notion. ◀

EXAMPLE 3

Paths in Collaboration Graphs In a collaboration graph, two people a and b are connected by a path when there is a sequence of people starting with a and ending with b such that the endpoints of each edge in the path are people who have collaborated. We will consider two particular collaboration graphs here. First, in the academic collaboration graph of people who have written papers in mathematics, the **Erdős number** of a person m (defined in terms of relations in Supplementary Exercise 14 in Chapter 9) is the length of the shortest path between m and the extremely prolific mathematician Paul Erdős (who died in 1996). That is, the Erdős number of a mathematician is the length of the shortest chain of mathematicians that begins with Paul Erdős and ends with this mathematician, where each adjacent pair of mathematicians have written a joint paper. The number of mathematicians with each Erdős number, last computed in 2006 as part of the Erdős Number Project, is shown in Table 1.

Links

Replace Kevin Bacon by your own favorite actor to invent a new party game.

In the Hollywood graph (see Example 3 in Section 10.1), two actors a and b are linked when there is a chain of actors linking a and b, where every two actors adjacent in the chain have acted in the same movie. In the Hollywood graph, the **Bacon number** of an actor c is defined to be the length of the shortest path connecting c and the well-known actor Kevin Bacon. As new movies are made, including new ones with Kevin Bacon, the Bacon number of actors can change. In Table 2 we show the number of actors with each Bacon number as of August 2017 using data from the Oracle of Bacon website. The origins of the Bacon number of an actor dates back to the early 1990s, when Kevin Bacon remarked that he had worked with everyone in Hollywood or someone who worked with them. This led some people to invent a party game where participants where challenged to find a sequence of movies leading from each actor named to Kevin Bacon. Before the web was born, you needed to be a movie expert to play this game well. Now, you need only consult the Oracle of Bacon. We can find a number similar to a Bacon number using any actor as the center of the acting universe. ◀

Koneksi Undirected Graf

Definition 3 An undirected graph is called *connected* if there is a path between every pair of distinct vertices of the graph. An undirected graph that is not *connected* is called *disconnected*. We say that we *disconnect* a graph when we remove vertices or edges, or both, to produce a disconnected subgraph.

Thus, any two computers in the network can communicate if and only if the graph of this network is connected.

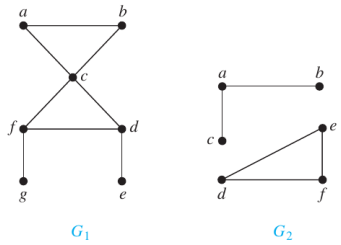


FIGURE 2 **The graphs G_1 and G_2.**

EXAMPLE 4 The graph G_1 in Figure 2 is connected, because for every pair of distinct vertices there is a path between them (the reader should verify this). However, the graph G_2 in Figure 2 is not connected. For instance, there is no path in G_2 between vertices a and d. ◀

THEOREM 1 There is a simple path between every pair of distinct vertices of a connected undirected graph.

Proof: Let u and v be two distinct vertices of the connected undirected graph $G = (V, E)$. Because G is connected, there is at least one path between u and v. Let $x_0, x_1, \ldots, x_n$, where $x_0 = u$ and $x_n = v$, be the vertex sequence of a path of least length. This path of least length is simple. To see this, suppose it is not simple. Then $x_i = x_j$ for some i and j with $0 \leq i < j$. This means that there is a path from u to v of shorter length with vertex sequence $x_0, x_1, \ldots, x_{i-1}, x_j, \ldots, x_n$ obtained by deleting the edges corresponding to the vertex sequence $x_i, \ldots, x_{j-1}$. ◁

Links

CONNECTED COMPONENTS A **connected component** of a graph G is a connected subgraph of G that is not a proper subgraph of another connected subgraph of G. That is, a connected component of a graph G is a maximal connected subgraph of G. A graph G that is not connected has two or more connected components that are disjoint and have G as their union.

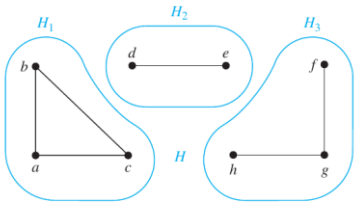


FIGURE 3 **The graph H and its connected components H_1, H_2, and H_3.**

EXAMPLE 5 What are the connected components of the graph H shown in Figure 3?

Solution: The graph H is the union of three disjoint connected subgraphs H_1, H_2, and H_3, shown in Figure 3. These three subgraphs are the connected components of H. ◀

EXAMPLE 6 **Connected Components of Call Graphs** Two vertices x and y are in the same component of a telephone call graph (see Example 4 in Section 10.1) when there is a sequence of telephone calls beginning at x and ending at y. When a call graph for telephone calls made during a particular day in the AT&T network was analyzed, this graph was found to have 53,767,087 vertices, more than 170 million edges, and more than 3.7 million connected components. Most of these components were small; approximately three-fourths consisted of two vertices representing pairs of telephone numbers that called only each other. This graph has one huge connected component with 44,989,297 vertices comprising more than 80% of the total. Furthermore, every vertex in this component can be linked to any other vertex by a chain of no more than 20 calls. ◀

Links

10.4.4 How Connected is a Graph?

Suppose that a graph represents a computer network. Knowing that this graph is connected tells us that any two computers on the network can communicate. However, we would also like to understand how reliable this network is. For instance, will it still be possible for all computers to communicate after a router or a communications link fails? To answer this and similar questions, we now develop some new concepts.

Sometimes the removal from a graph of a vertex and all incident edges produces a subgraph with more connected components. Such vertices are called **cut vertices** (or **articulation points**). The removal of a cut vertex from a connected graph produces a subgraph that is not connected. Analogously, an edge whose removal produces a graph with more connected components than in the original graph is called a **cut edge** or **bridge**. Note that in a graph representing a computer network, a cut vertex and a cut edge represent an essential router and an essential link that cannot fail for all computers to be able to communicate.

EXAMPLE 7 Find the cut vertices and cut edges in the graph G_1 shown in Figure 4.

Solution: The cut vertices of G_1 are b, c, and e. The removal of one of these vertices (and its adjacent edges) disconnects the graph. The cut edges are $\{a, b\}$ and $\{c, e\}$. Removing either one of these edges disconnects G_1. ◀

VERTEX CONNECTIVITY Not all graphs have cut vertices. For example, the complete graph K_n, where $n \geq 3$, has no cut vertices. When you remove a vertex from K_n and all edges incident to it, the resulting subgraph is the complete graph K_{n-1}, a connected graph. Connected graphs without cut vertices are called **nonseparable graphs**, and can be thought of as more connected than those with a cut vertex. We can extend this notion by defining a more granulated measure of graph connectivity based on the minimum number of vertices that can be removed to disconnect a graph.

A subset V' of the vertex set V of $G = (V, E)$ is a **vertex cut**, or **separating set**, if $G - V'$ is disconnected. For instance, in the graph in Figure 1, the set $\{b, c, e\}$ is a vertex cut with three vertices, as the reader should verify. We leave it to the reader (Exercise 51) to show that every connected graph, except a complete graph, has a vertex cut. We define the **vertex connectivity** of a noncomplete graph G, denoted by $\kappa(G)$, as the minimum number of vertices in a vertex cut.

κ is the lowercase Greek letter kappa.

When G is a complete graph, it has no vertex cuts, because removing any subset of its vertices and all incident edges still leaves a complete graph. Consequently, we cannot define $\kappa(G)$ as the minimum number of vertices in a vertex cut when G is complete. Instead, we set $\kappa(K_n) = n - 1$, the number of vertices needed to be removed to produce a graph with a single vertex.

Consequently, for every graph G, $\kappa(G)$ is minimum number of vertices that can be removed from G to either disconnect G or produce a graph with a single vertex. We have

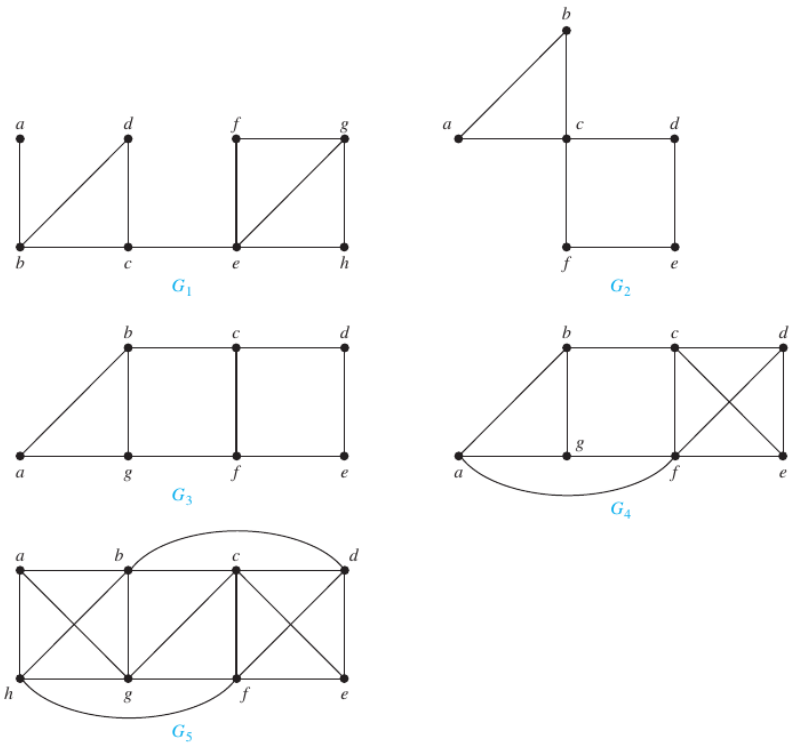


FIGURE 4 **Some connected graphs.**

$0 \leq \kappa(G) \leq n - 1$ if G has n vertices, $\kappa(G) = 0$ if and only if G is disconnected or $G = K_1$, and $\kappa(G) = n - 1$ if and only if G is complete [see Exercise 52(a)].

The larger $\kappa(G)$ is, the more connected we consider G to be. Disconnected graphs and K_1 have $\kappa(G) = 0$, connected graphs with cut vertices and K_2 have $\kappa(G) = 1$, graphs without cut vertices that can be disconnected by removing two vertices and K_3 have $\kappa(G) = 2$, and so on. We say that a graph is **k-connected** (or **k-vertex-connected**), if $\kappa(G) \geq k$. A graph G is 1-connected if it is connected and not a graph containing a single vertex; a graph is 2-connected, or **biconnected**, if it is nonseparable and has at least three vertices. Note that if G is a k-connected graph, then G is a j-connected graph for all j with $0 \leq j \leq k$.

EXAMPLE 8 Find the vertex connectivity for each of the graphs in Figure 4.

Solution: Each of the five graphs in Figure 4 is connected and has more than vertex, so each of these graphs has positive vertex connectivity. Because G_1 is a connected graph with a cut vertex, as shown in Example 7, we know that $\kappa(G_1) = 1$. Similarly, $\kappa(G_2) = 1$, because c is a cut vertex of G_2.

The reader should verify that G_3 has no cut vertices, but that $\{b, g\}$ is a vertex cut. Hence, $\kappa(G_3) = 2$. Similarly, because G_4 has a vertex cut of size two, $\{c, f\}$, but no cut vertices. It follows that $\kappa(G_4) = 2$. The reader can verify that G_5 has no vertex cut of size two, but $\{b, c, f\}$ is a vertex cut of G_5. Hence, $\kappa(G_5) = 3$. ◀

λ is the lowercase Greek letter lambda.

EDGE CONNECTIVITY We can also measure the connectivity of a connected graph $G = (V, E)$ in terms of the minimum number of edges that we can remove to disconnect it. If a graph has a cut edge, then we need only remove it to disconnect G. If G does not have a cut edge, we look for the smallest set of edges that can be removed to disconnect it. A set of edges E' is called an **edge cut** of G if the subgraph $G - E'$ is disconnected. The **edge connectivity** of a graph G, denoted by $\lambda(G)$, is the minimum number of edges in an edge cut of G. This defines $\lambda(G)$ for all connected graphs with more than one vertex because it is always possible to disconnect such a graph by removing all edges incident to one of its vertices. Note that $\lambda(G) = 0$ if G is not connected. We also specify that $\lambda(G) = 0$ if G is a graph consisting of a single vertex. It follows that if G is a graph with n vertices, then $0 \leq \lambda(G) \leq n - 1$. We leave it to the reader [Exercise 52(b)] to show that $\lambda(G) = n - 1$ where G is a graph with n vertices if and only if $G = K_n$, which is equivalent to the statement that $\lambda(G) \leq n - 2$ when G is not a complete graph.

EXAMPLE 9 Find the edge connectivity of each of the graphs in Figure 4.

Solution: Each of the five graphs in Figure 4 is connected and has more than one vertex, so we know that all of them have positive edge connectivity. As we saw in Example 7, G_1 has a cut edge, so $\lambda(G_1) = 1$.

The graph G_2 has no cut edges, as the reader should verify, but the removal of the two edges $\{a, b\}$ and $\{a, c\}$ disconnects it. Hence, $\lambda(G_2) = 2$. Similarly, $\lambda(G_3) = 2$, because G_3 has no cut edges, but the removal of the two edges $\{b, c\}$ and $\{f, g\}$ disconnects it.

The reader should verify that the removal of no two edges disconnects G_4, but the removal of the three edges $\{b, c\}$, $\{a, f\}$, and $\{f, g\}$ disconnects it. Hence, $\lambda(G_4) = 3$. Finally, the reader should verify that $\lambda(G_5) = 3$, because the removal of any two of its edges does not disconnect it, but the removal of $\{a, b\}$, $\{a, g\}$, and $\{a, h\}$ does. ◀

AN INEQUALITY FOR VERTEX CONNECTIVITY AND EDGE CONNECTIVITY

When $G = (V, E)$ is a noncomplete connected graph with at least three vertices, the minimum degree of a vertex of G is an upper bound for both the vertex connectivity of G and the edge connectivity of G. That is, $\kappa(G) \leq \min_{v \in V} \deg(v)$ and $\lambda(G) \leq \min_{v \in V} \deg(v)$. To see this, observe that deleting all the neighbors of a fixed vertex of minimum degree disconnects G, and deleting all the edges that have a fixed vertex of minimum degree as an endpoint disconnects G.

In Exercise 55, we ask the reader to show that $\kappa(G) \leq \lambda(G)$ when G is a connected noncomplete graph. Note also that $\kappa(K_n) = \lambda(K_n) = \min_{v \in V} \deg(v) = n - 1$ when n is a positive integer and that $\kappa(G) = \lambda(G) = 0$ when G is a disconnected graph. Putting these facts together establishes that for all graphs G,

$$\kappa(G) \leq \lambda(G) \leq \min_{v \in V} \deg(v).$$

Koneksi Directed Graf

Definition 4 A directed graph is *strongly connected* if there is a path from a to b and from b to a whenever a and b are vertices in the graph.

For a directed graph to be strongly connected there must be a sequence of directed edges from any vertex in the graph to any other vertex. A directed graph can fail to be strongly connected but still be in "one piece." Definition 5 makes this notion precise.

Definition 5 A directed graph is *weakly connected* if there is a path between every two vertices in the underlying undirected graph.

That is, a directed graph is weakly connected if and only if there is always a path between two vertices when the directions of the edges are disregarded. Clearly, any strongly connected directed graph is also weakly connected.

EXAMPLE 10 Are the directed graphs G and H shown in Figure 5 strongly connected? Are they weakly connected?

Solution: G is strongly connected because there is a path between any two vertices in this directed graph (the reader should verify this). Hence, G is also weakly connected. The graph H is not strongly connected. There is no directed path from a to b in this graph. However, H is weakly connected, because there is a path between any two vertices in the underlying undirected graph of H (the reader should verify this). ◀

STRONG COMPONENTS OF A DIRECTED GRAPH The subgraphs of a directed graph G that are strongly connected but not contained in larger strongly connected subgraphs, that is, the maximal strongly connected subgraphs, are called the **strongly connected components** or **strong components** of G. Note that if a and b are two vertices in a directed graph, their strong components are either the same or disjoint. (We leave the proof of this last fact as Exercise 17.)

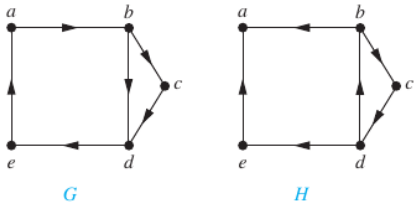


FIGURE 5 **The directed graphs G and H.**

EXAMPLE 11 The graph H in Figure 5 has three strongly connected components, consisting of the vertex a; the vertex e; and the subgraph consisting of the vertices b, c, and d and edges (b, c), (c, d), and (d, b). ◀

EXAMPLE 12 **The Strongly Connected Components of the Web Graph** The web graph introduced in Example 5 of Section 10.1 represents web pages with vertices, and links with directed edges. Determining the size of the web graph is a challenging problem because it is extraordinarily large. A snapshot of the web in 1999 produced a web graph with over 200 million vertices and over 1.5 billion edges. In 2010 the web graph was estimated to have at least 55 billion vertices and one trillion edges. From these limited data (and from what is readily apparent if you have used the web in recent years), the number of web pages has a rapid rate of growth. (See [Br00] and the web resources for details.)

Links

In 2010 it was estimated that 40 terrabytes of computer memory would be needed to represent the adjacency matrix of the web graph, but the storage requirements today are vastly larger.

The underlying undirected graph of the web graph, as it existed in 1999, is not connected, but it has a connected component that includes approximately 90% of the vertices in the graph. The subgraph of the original directed graph corresponding to this connected component of the underlying undirected graph (that is, with the same vertices and all directed edges connecting vertices in this subgraph) has one very large strongly connected component and many small ones. The former is called the **giant strongly connected component (GSCC)** of the directed graph. A web page in this component can be reached following links starting at any other page in this component. The GSCC in the web graph was found to have over 53 million vertices. The remaining vertices in the large connected component of the undirected graph represent three different types of web pages: pages that can be reached from a page in the GSCC, but do not link back to these pages following a series of links; pages that link back to pages in the GSCC following a series of links, but cannot be reached by following links on pages in the GSCC; and pages that cannot reach pages in the GSCC and cannot be reached from pages in the GSCC following a series of links. Each of these three other sets was found to have approximately 44 million vertices. (It is rather surprising that these three sets are close to the same size.) ◀

Tutorial Praktikum

Tutorial01.py

Mahasiswa dapat menjalankan skrip Python di bawah ini untuk memahami konsep-konsep dasar connectivity secara visual. Kode ini menunjukkan bagaimana mendefinisikan path dan circuit, serta bagaimana NetworkX menganalisis keterhubungan, komponen terhubung, cut vertex, dan cut edge.

Task Praktikum

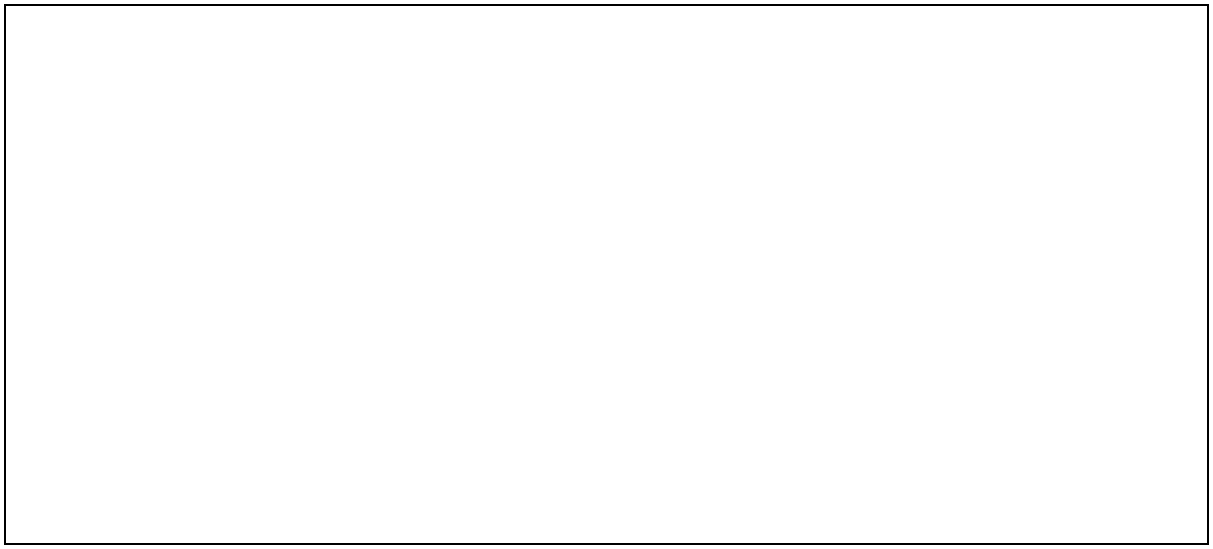
Pada praktikum matdis lanjut mahasiswa boleh menggunakan tools AI, dokumentasi official dan lain-lain, sesuai RPS tidak semua tugas akan dinilai, namun akan digunakan sebagai landasan untuk mengerjakan pekerjaan selanjutnya / modul selanjutnya.

Task Individu

Task01_Individu.py

1. Mahasiswa diminta untuk menjalankan dan menganalisis kode pada modul_praktikum_minggu_3 tutorial.py
2. Tugas individu ini bertujuan untuk memastikan setiap mahasiswa memahami dasar-dasar konektivitas sebelum masuk ke studi kasus kolaboratif.

Tuliskan Lesson learnt task ini (boleh screenshoot hasil akhir)



Task Kelompok (Soal Kolaboratif & Sharing)

Setiap anggota kelompok akan memilih satu dari tiga studi kasus (soal 2, 3, atau 4) untuk dikerjakan. Setelah selesai, setiap mahasiswa akan mempresentasikan hasil analisisnya kepada kelompok.

Setelah semua anggota selesai mengerjakan, setiap mahasiswa harus:

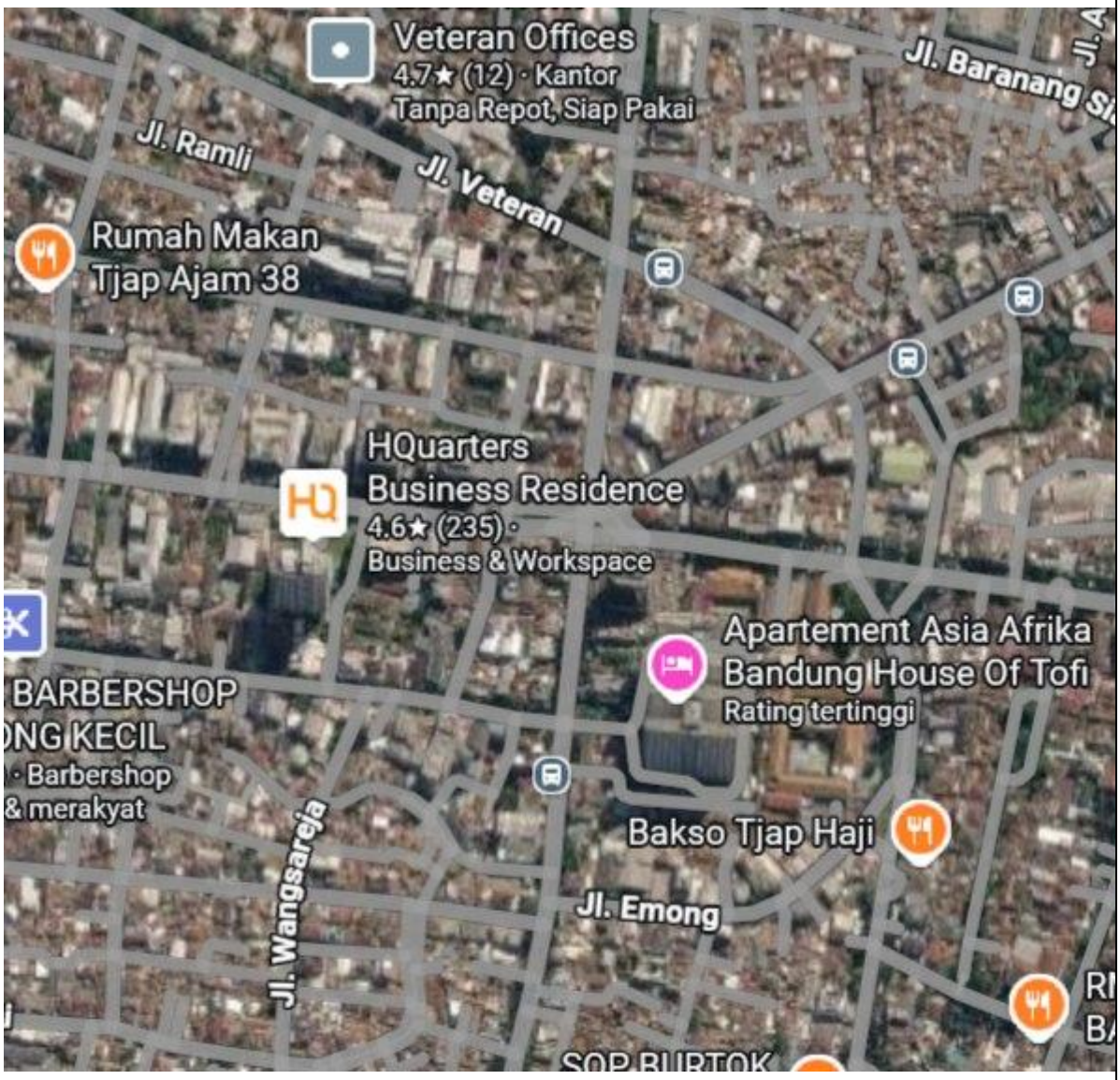
1. Menjelaskan hasil analisis dan kode programnya kepada anggota kelompok lain.
2. Membuat lesson learnt tentang apa yang dipelajari dari studi kasus teman-teman mereka.

Task02_ Analisis Jaringan Jalan Raya dan Kemacetan.py



- **Skenario**: Anda adalah seorang analis transportasi di Bandung. Anda ingin memodelkan kemacetan di persimpangan jalan Soekarno-Hatta (Simpang Lima). merupakan persimpangan terbanyak paling populer di Kota Bandung. Simpang ini adalah pertemuan lima jalan, yaitu Jl. Asia Afrika, Jl. Sunda, Jl. Sunda–Karapitan, Jl. Gatot Subroto, dan Jl. Ahmad Yani. . Untuk mempermudah, asumsikan persimpangan ini adalah sebuah graf, di mana simpul adalah persimpangan dan sisi adalah jalan.
- **Tugas**:
    - Search simpang lima di google maps (print screen simpang lima)
    - Modelkan persimpangan ini sebagai graf tak berarah.
    - Identifikasi `Cut Vertex` dan `Cut Edge` pada graf tersebut.
    - Jelaskan makna `Cut Vertex` dan `Cut Edge` dalam konteks jaringan jalan raya di persimpangan tersebut. Mengapa persimpangan tersebut sering mengalami kemacetan? Kaitkan dengan konsep `Cut Vertex`.
    - Visualisasikan graf, lalu visualisasikan kembali setelah salah satu `Cut Vertex` dihapus untuk menunjukkan bahwa graf menjadi tidak terhubung.

Lesson learnt:
- Memahami konsep cut vertex dan cut edge dalam analisis jaringan, yaitu simpul atau sisi yang apabila dihapus dapat meningkatkan jumlah komponen graf yang terhubung.
- Menganalisis hubungan antara struktur graf dan kemacetan lalu lintas. Persimpangan yang menjadi titik penghubung utama dapat menjadi titik kritis ketika tidak tersedia jalur alternatif.
- Melatih kemampuan memodelkan permasalahan nyata ke dalam struktur data graf sebagai dasar analisis konektivitas dan pencarian jalur alternatif.

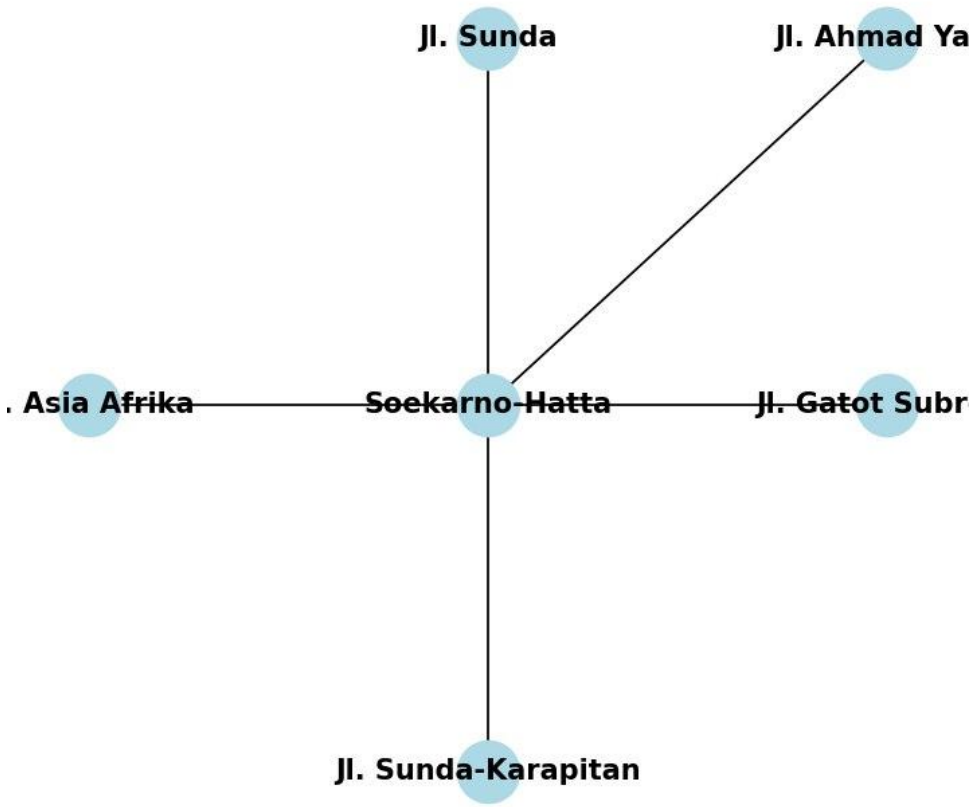


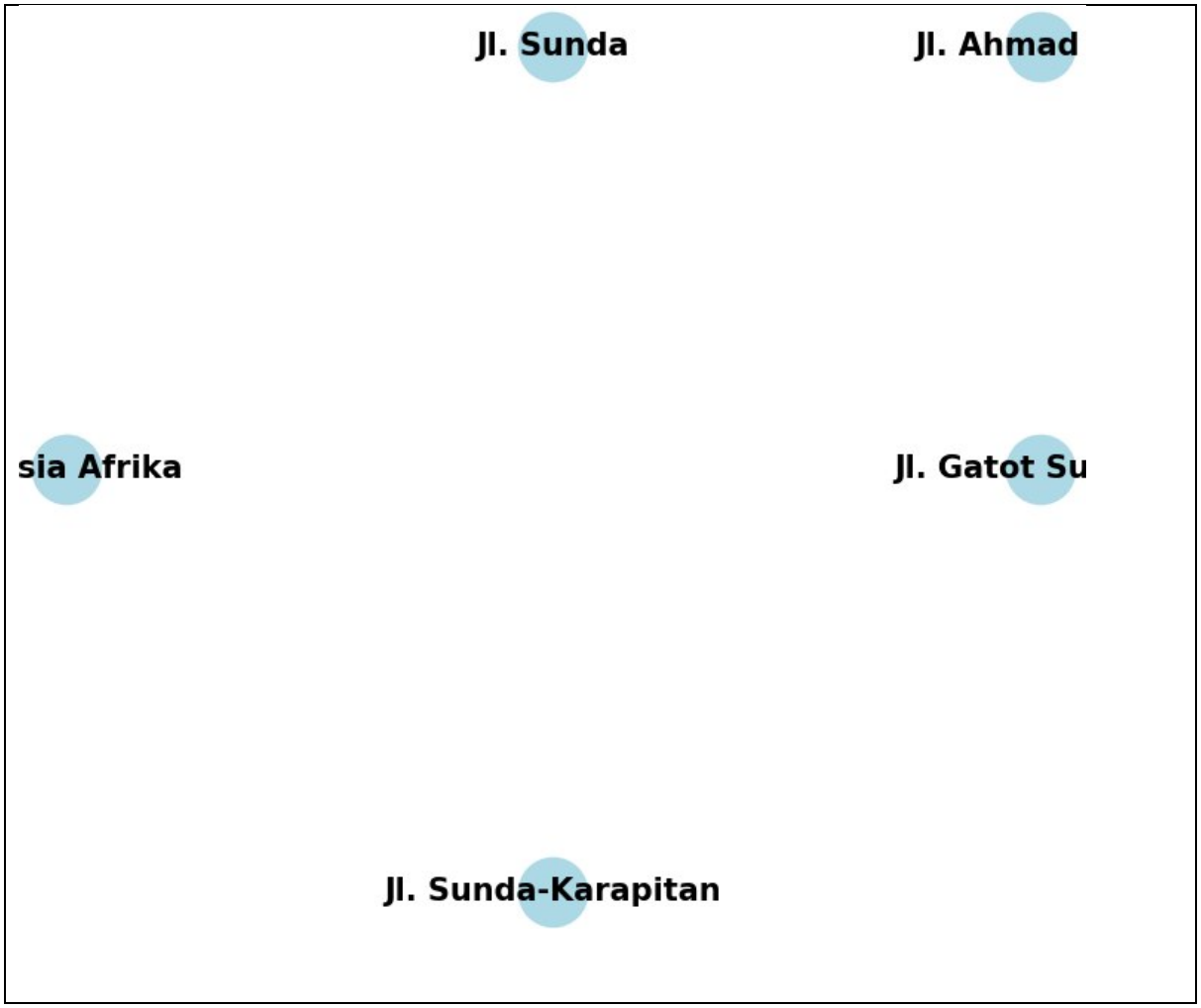
Cut vertex: a (Simpang 5 Soekarno-Hatta)
Cut edge: ada pada tiap jalan yang menguhubungkan vertexnya

Anggap satu simpangan hanya berisi 5 jalan besar. Jika simpang lima ini ditutup, maka seluruh jaringan tidak bisa saling melewati (terblock). Mobilitas akan lumpuh karena tidak ada jalur alternatif. Itulah Cut vertex yang dimaksud disini.

Kemudian karena hanya terhubung oleh satu jalan pada tiap vertex, sehingga saat jalan itu dihalangi/dihilangkan, maka arah yang akan dari jalan tersebut maupun dari luar jalan

tersebut tidak akan bisa lewat. Namun, jalan lain yang masih terhubung dengan simpangan tersebut akan tetap bisa saling mengakses.





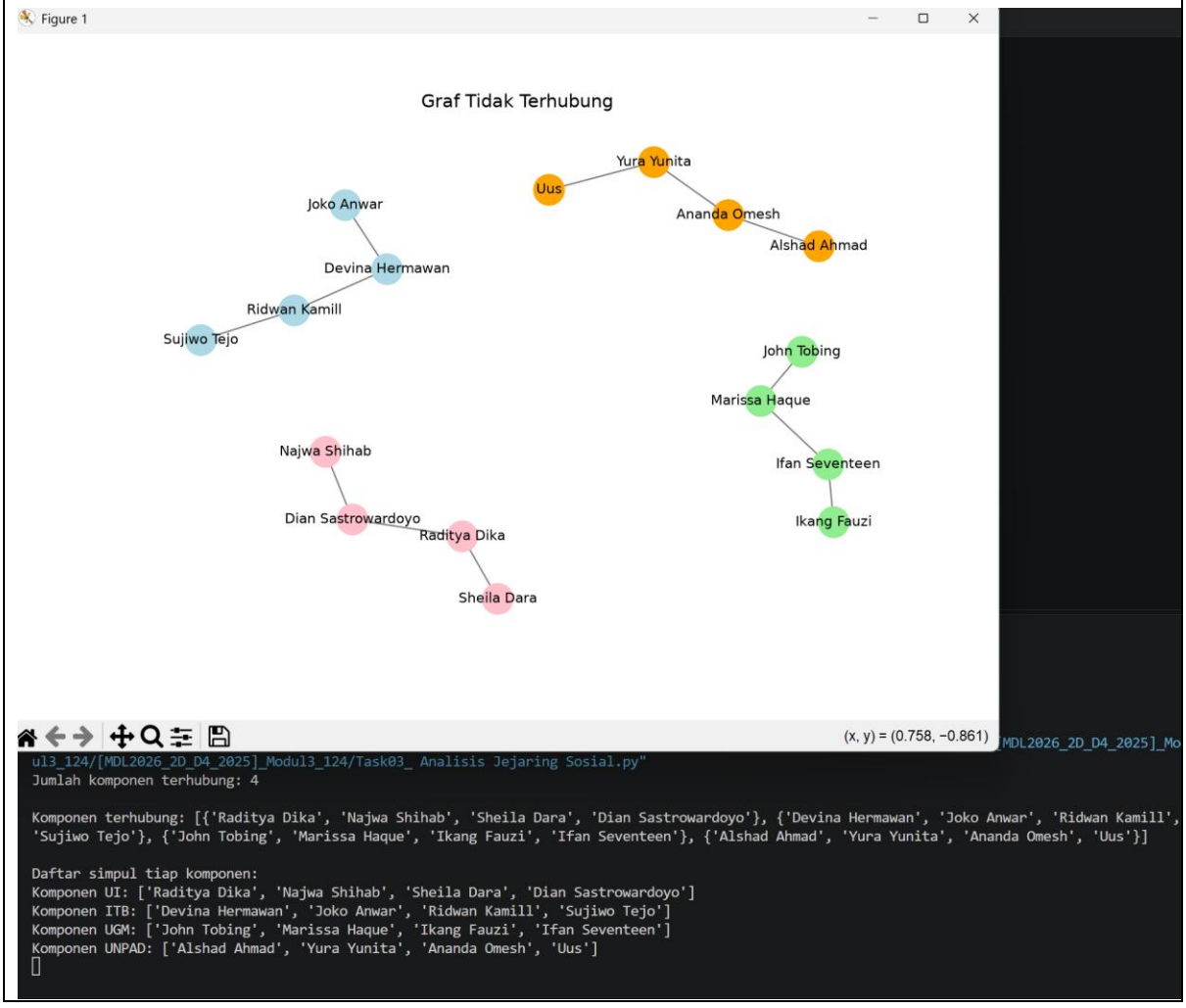
Task03_ Analisis Jejaring Sosial.py

- **Skenario**: Anda mengamati jejaring sosial antara para tokoh publik, di mana pertemanan diwakili oleh simpul, dan afiliasi dengan universitas (UGM, ITB, UI, Unpad) membentuk kelompok-kelompok. Tokoh-tokoh ini tidak selalu saling mengenal, sehingga jejaring ini memiliki beberapa connected components.
- Untuk membuat graf, cari dari wikipedia tokoh masyarakat yang merupakan lulusan dari universitas tersebut baik S1, S2 maupun S3, masing-masing universitas 3 orang (total 12 vertex)
- **Tugas**:
    - Buatlah sebuah graf tak berarah yang memiliki setidaknya tiga connected components. Setiap komponen harus merepresentasikan sebuah afiliasi universitas dan memiliki minimal 3 simpul (tokoh).
    - Tampilkan jumlah connected components dan daftar simpul di setiap komponen.
    - Jelaskan makna connected components dalam konteks jejaring sosial ini. Kaitkan dengan asumsi bahwa pertemanan cenderung terjadi di dalam lingkungan afiliasi yang sama.
    - Visualisasikan graf Anda, lalu gunakan warna yang berbeda untuk setiap connected component untuk memperjelas hasilnya.

Connected component adalah kumpulan simpul yang saling terhubung lewat suatu path, tapi tidak punya hubungan ke simpul di luar kumpulan itu. Di jejaring sosial ini, satu komponen berarti satu lingkar pertemanan. Dua tokoh di komponen yang sama bisa saling terjangkau, langsung atau lewat perantara. Contohnya, Najwa Shihab terhubung ke Sheila Dara lewat Dian Sastrowardoyo dan Raditya Dika. Sebaliknya, tokoh dari komponen yang berbeda, misalnya Najwa Shihab dan Joko Anwar, tidak punya jalur sama sekali, jadi dalam graf ini tidak ada hubungan di antara mereka.

Graf yang dibuat punya 4 connected components: UI, ITB, UGM, dan Unpad. Masing-masing berisi 4 tokoh, total 16 simpul. Tidak ada sisi yang menghubungkan tokoh dari universitas berbeda, jadi tiap universitas membentuk satu komponen sendiri. Ini sesuai asumsi bahwa pertemanan cenderung terjadi di lingkungan afiliasi yang sama, karena hubungan hanya ada di dalam satu kampus dan graf akhirnya terpecah jadi 4 kelompok.

Lesson Learnt
Dari tugas ini aku memahami bahwa connected component adalah kumpulan simpul yang saling terjangkau lewat suatu path dan tidak punya hubungan ke simpul di luarnya, sehingga bisa dipakai untuk membaca kelompok yang terpisah dalam sebuah jaringan.



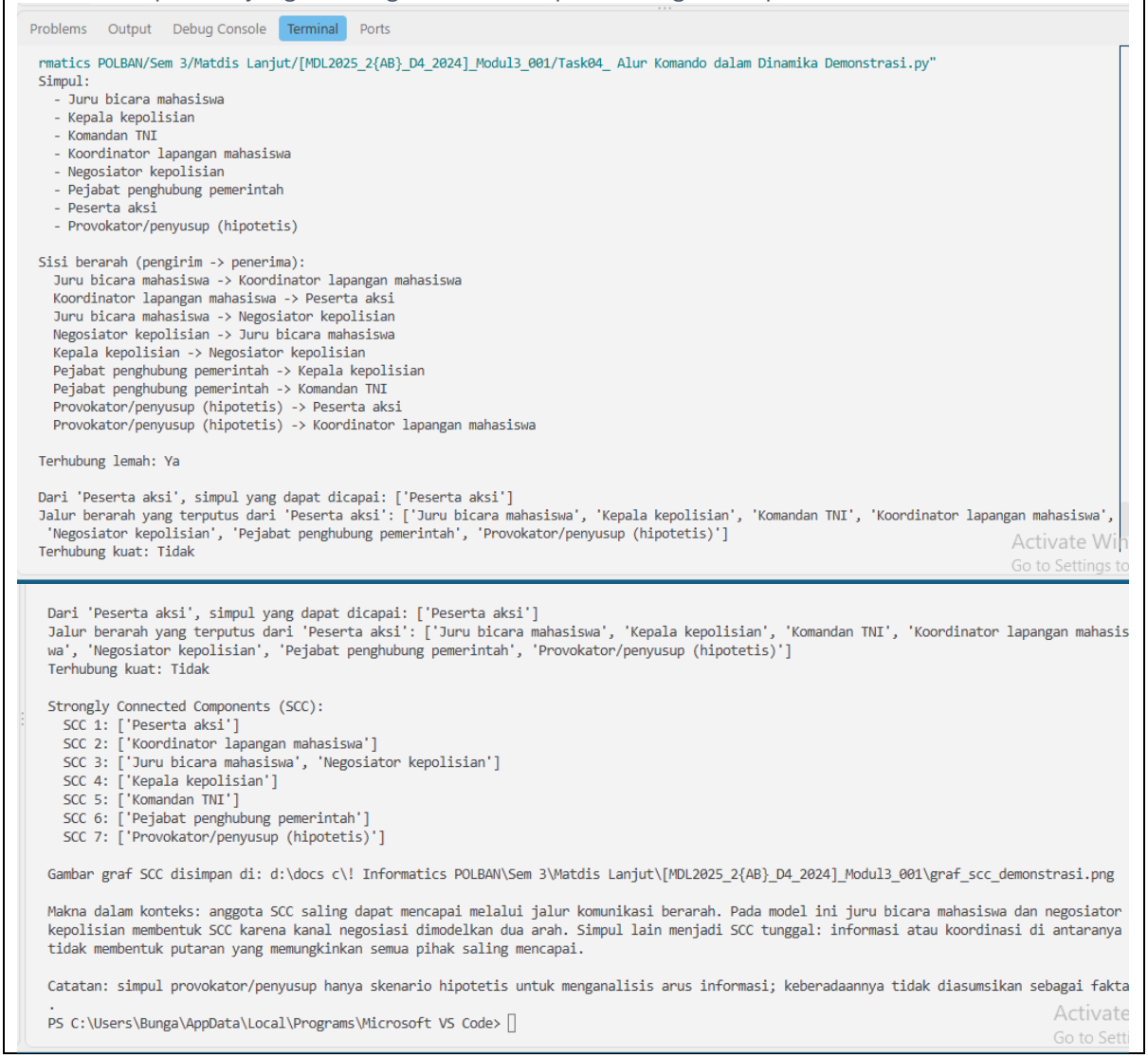
Task04_ Alur Komando dalam Dinamika Demonstrasi.py

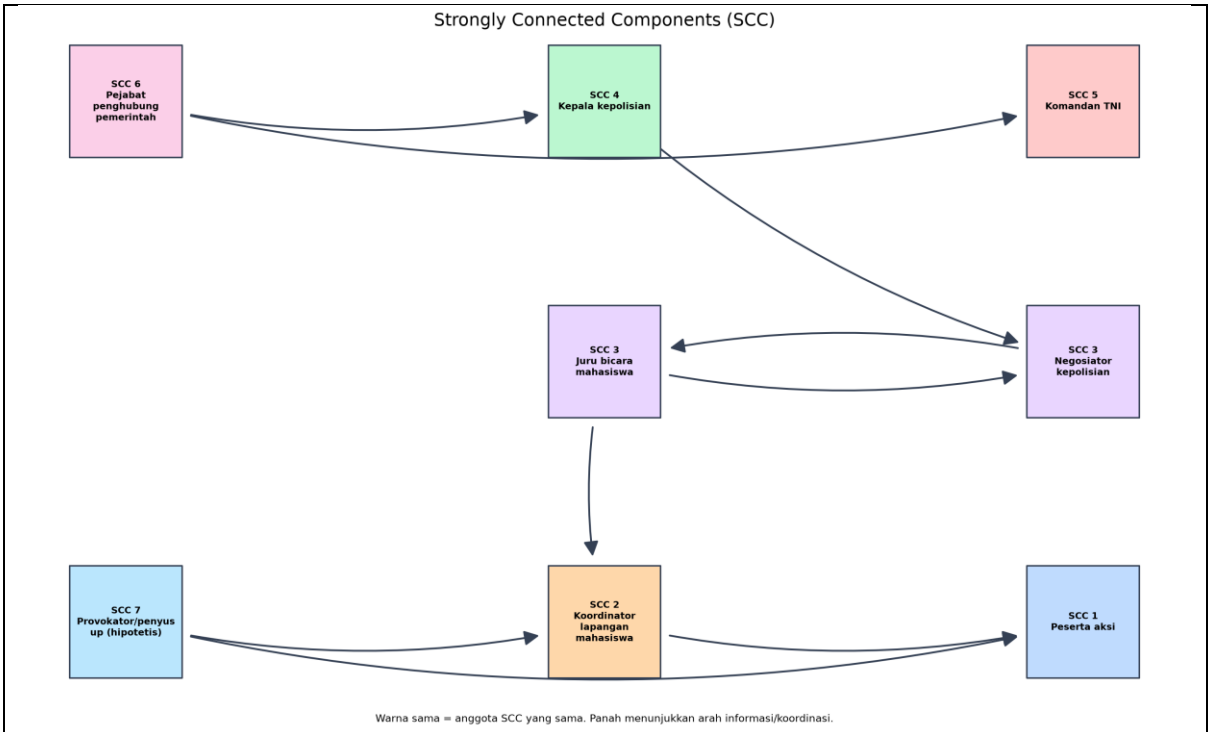
- **Skenario**: Anda menganalisis alur komando dan informasi dalam sebuah demonstrasi yang melibatkan berbagai pihak, yaitu Mahasiswa, Polisi, TNI, dan Pemerintah. Aliran

informasi dan perintah dalam skenario ini bersifat satu arah dan tidak selalu timbal balik.

- Sebelum mengerjakan tugas, pahami domain masalah bagaimana realnya komunikasi saat adanya demonstrasi dari pihak mahasiswa (ada aktor siapa saja), polisi, tni dan pemerintah, misal tambahkan penyusup (demo bayaran) dan lain-lain.
- **Tugas**:
    - Modelkan skenario ini sebagai Directed Graph. Tentukan simpul-simpulnya (misalnya, Juru Bicara Mahasiswa, Kepala Polisi, Komandan TNI, Juru Bicara Pemerintah).
    - Buatlah graf yang terhubung lemah tetapi tidak terhubung kuat.
    - Buktikan bahwa graf tersebut tidak terhubung kuat dengan mengidentifikasi jalur komando yang terputus.
    - Temukan dan tampilkan `Strongly Connected Components` dari graf tersebut.
    - Jelaskan makna `Strongly Connected Components` dalam konteks skenario ini.

Makna dalam konteks: anggota SCC saling dapat mencapai melalui jalur komunikasi berarah. Pada model ini juru bicara mahasiswa dan negosiator kepolisian membentuk SCC karena kanal negosiasi dimodelkan dua arah. Simpul lain menjadi SCC tunggal: informasi atau koordinasi di antaranya tidak membentuk putaran yang memungkinkan semua pihak saling mencapai.





Lesson Learnt:
Saya belajar bahwa pemodelan sistem kompleks menggunakan graf berarah sangat efektif untuk memetakan alur komunikasi dalam skenario demonstrasi. Analisis membuktikan jaringan ini terhubung lemah namun tidak terhubung kuat karena aliran informasi tidak selalu timbal balik, terlihat dari "Peserta aksi" yang hanya bertindak sebagai penerima akhir pesan tanpa jalur respons ke atas. Konsep *Strongly Connected Components* (SCC) terbukti berguna untuk menyoroti titik negosiasi dan komunikasi dua arah secara spesifik, seperti antara juru bicara mahasiswa dan negosiator kepolisian. Selain itu, pemodelan ini mampu memfasilitasi simulasi intervensi pihak ketiga (provokator) untuk melihat jalur injeksi informasi satu arah, serta membuktikan bahwa perhitungan status keterhubungan graf dapat diautomatisasi secara efisien menggunakan pustaka NetworkX.

Use the "Insert Citation" button to add citations to this document.